V. SUBIKSHA
192424135
JAVA

Q1. Design a class Hierarchy (Pseudocode)

```
START
Create class person
    Declare name
    Declare age
    Method getPersonDetails ()
        Input name
        Input age
    End Method
End class

Create class Student extends person
    Declare registerNumber
    Declare marks [5]
    Declare average
    Method getStudentDetails ()
        Input registerNumber
        FOR i=0 to 4
            Input mark [i]
        END FOR
    END method
    Method CalculateAverage ()
        Sum = 0
        FOR i=0 to 4
            Sum = sum + marks[i]
        END FOR
```

average = sum/5
Return average

END class

Create class ResearchStudent extends Student
Declare researchArea

method
displayCompleteDetails ()

Display name
Display age
Display registerNumber
Display marks
Display average
Display researchArea

End class

Main method

Create object RS of Research Student

call RS.getpersonDetails ()
call RS.getStudentDetails ()

Input researchArea

call. RS. calculateAverage ()

call RS. displayCompleteDetails ()

STOP

Q2. Complete the Missing Pseudocode

```
BEGIN

READ N
READ ARRAY A

FOR i ← 0 TO N-1
  FOR j ← i+1 TO N-1
     IF A[i] == A[j]
        DELETE A[j]
     ENDIF
  ENDFOR
ENDFOR

PRINT Modified Array

END
```

Q3. Debug the Pseudocode

Incorrect line:

RETURN length * breadth

correct

```
METHOD area (length)
RETURN length * length
END METHOD

METHOD area (length, breadth)
RETURN length * breadth
END METHOD
```

Q4 Trace the Execution

sum = 0

i = 1 → odd → sum = 0 - 1 = -1

i = 2 → even → sum = -1 + 2 = 1

i = 3 → odd → sum = 1 - 3 = -2

i = 4 → even → sum = -2 + 4 = 2

i = 5 → odd → sum = 2 - 5 = -3

output

-3

Q5. Efficient pseudocode

```
BEGIN

READ N
READ ARRAY A
largest ← -∞
Largest ← -∞
FOR i ← 0 TO N-1
    IF A[i] > largest THEN
        second ← largest
        largest ← A[i]
  ELSE IF A[i] > second AND A[i] != largest THEN
        second ← A[i]
  ENDIF
ENDFOR
PRINT second
```

Q6. Interface

```
INTERFACE payment
   METHOD pay(amount)
END INTERFACE

CLASS CreditCard IMPLEMENTS payment
   METHOD pay (amount)
      PRINT "Paid using Credit Card"
   END METHOD
END CLASS

CLASS UPI IMPLEMENTS payment
   METHOD pay (amount)
      PRINT "paid using UPI"
   END METHOD
END CLASS

CLASS NetBanking IMPLEMENTS Payment
   METHOD pay (amount)
      PRINT "Paid using Net Banking"
   END METHOD
END CLASS

BEGIN
READ choice
READ amount
IF choice = 1 THEN
   P<- NEW CreditCard
ELSE IF choice = 2 THEN
   P<-NEW UPI
ELSE
   P<- NEW NetBanking
ENDIF
P.pay(amount)
END
```

Q7. Runtime polymorphism

Fill the blanks

```
CLASS Dog EXTENDS Animal
Animal obj ← NEW Dog()
```

Complete

```
CLASS Animal
  METHOD speak()
    PRINT "Animal"
  END METHOD
END CLASS

CLASS Dog Extends Animal
  METHOD speak()
    PRINT "Barks"
  END METHOD
END CLASS

Animal obj ← NEW Dog()
obj.speak()
```

Q8. Correct constructor:

```
CLASS Employee

DATA
  id
  name

CONSTRUCTOR(id, name)
  this.id ← id
  this.name ← name

END CONSTRUCTOR

END CLASS
```

Q9.

```
CLASS vehicle
  vehicleNo
  ownerName

  METHOD display()
    PRINT vehicleNo,
    ownerName
END CLASS

CLASS Car EXTENDS vehicle
  fuelType
END CLASS

CLASS Truck EXTENDS vehicle
  loadCapacity
END CLASS

BEGIN
READ N

FOR i ← 1 TO N
  READ type

IF type = "car"
  obj ← NEW Car

ELSE
  obj ← NEW Truck

ENDIF
  READ vehicleNo,

ownerName
  STORE obj

ENDFOR

FOR EACH obj
  obj.display()
ENDFOR

END
```